

Stage	Topics Covered
Getting Started: Basics of C#	Hello World, Variables, Data Types, Control Flow (if, else, switch), Loops
Understanding OOP	Classes, Objects, Inheritance, Polymorphism, Encapsulation, Abstraction, Constructors, Destructors
Deepening Knowledge in C# Fundamentals	Exception Handling, Collections, LINQ, Delegates and Events, Namespaces, Assemblies
Advanced C# Features	Asynchronous Programming (<code>async</code> and <code>await</code>), Advanced LINQ, Generics, Reflection, Nullable Types
Working with Files and Streams	Reading/Writing Files, Streams, Serialization and Deserialization
Database Interaction	Connecting to databases, Executing queries, CRUD operations, Entity Framework or ADO.NET
Advanced Topics	SOLID and Design Patterns

Resource:

Beginner to Intermediate:

1. Online Platforms:

- Codecademy - Learn C#
- Pluralsight - C# Fundamentals with Visual Studio
- edX - Introduction to C#

2. Books:

- "C# 9.0 in a Nutshell" by Joseph Albahari and Ben Albahari
- "Head First C#" by Andrew Stellman and Jennifer Greene
- "C# Programming Yellow Book" by Rob Miles

3. Interactive Learning:

- Microsoft Learn - C#
- [CodeCombat - Learn C#](#)

4. Video Tutorials:

- The Net Ninja - C# Tutorial for Beginners
- Programming with Mosh - C# Basics for Beginners

Intermediate to Advanced:

1. Online Courses:

- Pluralsight - Advanced C# Programming
- Udemy - C# Advanced Topics: Take Your C# Skills to the Next Level
- LinkedIn Learning - C# Tips and Traps

2. Books:

- "Effective C#: 50 Specific Ways to Improve Your C#" by Bill Wagner
- "CLR via C#" by Jeffrey Richter
- "C# in Depth" by Jon Skeet

3. Coding Challenges:

- HackerRank - C# Practice
- LeetCode - C#

4. Advanced Topics and Specialized Learning:

- Microsoft Docs - Asynchronous programming in C#
- Pluralsight - Design Patterns in C#
- Coursera - .NET Design Patterns for Object-Oriented Software